

$$1. P(H) = \frac{1}{3} \Rightarrow P(T) = 1 - P(H) = 1 - \frac{1}{3} = \boxed{\frac{2}{3}}$$

$$2. P(\text{Red}) = \frac{26}{52}$$

$$P(\text{Face}) = \frac{12}{52}$$

$$P(\text{Red AND Face}) = \frac{6}{52}$$

$$P(\text{Red OR Face}) = P(\text{Red}) + P(\text{Face}) - P(\text{Red AND Face})$$

$$= \frac{26}{52} + \frac{12}{52} - \frac{6}{52} = \frac{26 + 12 - 6}{52}$$

$$= \frac{32}{52} = \frac{8}{13}$$

$$3. \boxed{95\%}$$

$$4. P(\text{Spade AND Black}) = \frac{13}{52}$$

$$P(\text{Black}) = \frac{26}{52}$$

$$P(\text{Spade} | \text{Black}) = \frac{P(\text{Spade AND Black})}{P(\text{Black})} = \frac{13/52}{26/52} = \boxed{\frac{1}{2}}$$

5. ~~ANMR~~

$D \in \{0, 1\}$ True damage

$P \in \{0, 1\}$ Predicted damage

Prevalence: $IP(D=1) = \frac{1}{100} = 0.01$

FPR: $IP(P=1 | D=0) = \frac{1}{50} = 0.02$

FNR: $IP(P=0 | D=1) = 0$

D) Only 33.6% of the signs predicted damaged are actually damaged. That means most signs predicted damaged are actually fine, which could lead to wasted resources. In spite of the perfect TPR, I would not recommend NYC invest in this technology.

A) TPR: $IP(P=1 | D=1) = 1 - IP(P=0 | D=1) = \boxed{1}$

B) $IP(P=1) = \underbrace{IP(P=1 | D=0)}_{\frac{1}{50}} \underbrace{IP(D=0)}_{1 - \frac{1}{100} = \frac{99}{100}} + \underbrace{IP(P=1 | D=1)}_1 \underbrace{IP(D=1)}_{\frac{1}{100}} = \frac{1}{50} \cdot \frac{99}{100} + 1 \cdot \frac{1}{100} = \frac{99}{5000} + \frac{1}{100} = \frac{99}{5000} + \frac{50}{5000} = \frac{149}{5000} = 0.0298$

C) $IP(D=1 | P=1) = \frac{IP(P=1 | D=1) IP(D=1)}{IP(P=1)}$

$$= \frac{1 \cdot \frac{1}{100}}{\frac{149}{5000}} = \frac{1}{100} \cdot \frac{5000}{149} = \frac{5000}{14900} = \boxed{0.3356}$$

$$\boxed{33.6\%}$$